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DISCOVERING LIVE HOSTS USING NMAP SCANS

Enumerating Targets:

Task 4 Enumerating Targets

We mentioned the different techniques we can use for scanning in Task 1. Before we explain each in detail and test it against a live target, we need to specify the targets we want to scan. Generally speaking, you can provide a list, a range, or a subnet. Examples of target specification are:

- list: `MACHINE_IP scanme.nmap.org example.com` will scan 3 IP addresses.
- range: `10.11.12.15-20` will scan 6 IP addresses: `10.11.12.15`, `10.11.12.16`, ..., and `10.11.12.20`.
- subnet: `MACHINE_IP/30` will scan 4 IP addresses.

You can also provide a file as input for your list of targets, `nmap -iL list_of_hosts.txt`.

If you want to check the list of hosts that Nmap will scan, you can use `nmap -sL TARGETS`. This option gives you a detailed list of hosts that Nmap will scan without actually scanning them; however, Nmap will attempt reverse-DNS resolution on all targets to obtain their names. Names might reveal various information to the pentester. (If you don't want Nmap to use the DNS server, you can add `-n`.)

Connecting to the Network

In this room, you will work with a small network consisting of one Windows host and one Ubuntu host to practice identifying live systems on a network.

When you open the room, you will see a network diagram showing the available hosts. To begin, click the **Start** button to deploy the network. Allow up to 2 minutes for all machines to initialise properly. Once the network has finished loading, launch the AttackBox by clicking the **Start AttackBox** button. When the AttackBox opens in split view, open a Terminal and run `Nmap` to perform host discovery, then answer the questions in this room.

Answer the questions below

What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?

✓ Correct Answer

How many IP addresses will Nmap scan if you provide the following range: 10.10.0-255.101-125?

✓ Correct Answer

ARP SCAN:

The source MAC address belongs to the AttackBox, while the destination MAC address is the broadcast address. This is expected, as the target host's MAC address is not yet known. However, the target IP address being queried is visible in the Info column. As shown in the figure, ARP requests are sent sequentially to each IP address on the subnet, starting from `10.201.24.1`.

When a host with the queried IP address is online, it responds with an ARP reply containing its MAC address. Receiving this reply confirms that the host is active and reachable on the network.

ARP scanning is particularly useful during post-exploitation and internal network enumeration. Once an attacker gains access to a system within a network, ARP scans can be used to quickly and reliably identify other live hosts on the same local segment. Because ARP operates at Layer 2 and is often not filtered by firewalls, it is an effective tool for red-team operations and internal penetration testing.

Answer the questions below

How many hosts are found alive after scanning the CONNECTION_IP/24?

✓ Correct Answer

ICMP SCAN:

Although we didn't receive any reply and couldn't determine which hosts were online, it is essential to note that this scan sent ICMP address-mask requests to every valid IP address and waited for a reply. Each ICMP request was sent twice, as shown in the screenshot below.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.200.5.2	10.200.6.1	ICMP	32	Address mask request
2	0.000033951	10.200.5.2	10.200.6.2	ICMP	32	Address mask request
3	0.000044881	10.200.5.2	10.200.6.3	ICMP	32	Address mask request
4	0.000055022	10.200.5.2	10.200.6.4	ICMP	32	Address mask request
5	0.000064652	10.200.5.2	10.200.6.5	ICMP	32	Address mask request
6	0.000074092	10.200.5.2	10.200.6.6	ICMP	32	Address mask request
7	0.000083532	10.200.5.2	10.200.6.7	ICMP	32	Address mask request
8	0.000093712	10.200.5.2	10.200.6.8	ICMP	32	Address mask request
9	0.000104403	10.200.5.2	10.200.6.9	ICMP	32	Address mask request
10	0.000114613	10.200.5.2	10.200.6.10	ICMP	32	Address mask request
11	1.001146808	10.200.5.2	10.200.6.1	ICMP	32	Address mask request
12	1.001192019	10.200.5.2	10.200.6.2	ICMP	32	Address mask request
13	1.001208509	10.200.5.2	10.200.6.3	ICMP	32	Address mask request
14	1.001218390	10.200.5.2	10.200.6.4	ICMP	32	Address mask request
15	1.001228180	10.200.5.2	10.200.6.5	ICMP	32	Address mask request
16	1.001237880	10.200.5.2	10.200.6.6	ICMP	32	Address mask request
17	1.001247560	10.200.5.2	10.200.6.7	ICMP	32	Address mask request
18	1.001258001	10.200.5.2	10.200.6.8	ICMP	32	Address mask request
19	1.001268871	10.200.5.2	10.200.6.9	ICMP	32	Address mask request
20	1.001278851	10.200.5.2	10.200.6.10	ICMP	32	Address mask request

Frame 1: 31... 0000 45 00 00 20 d1 88 00 00 30 01 90 c2 0a c8 05 02 E... 8...
Raw packet: 0010 0a c8 06 01 11 00 ea ee 04 11 00 00 00 00 00 00

Answer the questions below

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

-PP

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

-PM

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

-PE

✓ Correct Answer

TCP AND UDP SCAN:

No.	Time	Source	Destination	Protocol	Length	Info
128	4.868785798	10.200.5.2	10.200.6.96	UDP	28	53835 → 40125 Len=0
129	4.868806599	10.200.5.2	10.200.6.97	UDP	28	53835 → 40125 Len=0
130	4.868810369	10.200.5.2	10.200.6.98	UDP	28	53835 → 40125 Len=0
131	4.868838750	10.200.5.2	10.200.6.101	UDP	28	53835 → 40125 Len=0
132	4.868864050	10.200.5.2	10.200.6.102	UDP	28	53835 → 40125 Len=0
133	4.868883441	10.200.5.2	10.200.6.104	UDP	28	53835 → 40125 Len=0
134	4.868898091	10.200.5.2	10.200.6.105	UDP	28	53835 → 40125 Len=0
135	4.868917402	10.200.5.2	10.200.6.107	UDP	28	53835 → 40125 Len=0
136	4.868949623	10.200.5.2	10.200.6.109	UDP	28	53835 → 40125 Len=0
137	4.868969793	10.200.5.2	10.200.6.111	UDP	28	53835 → 40125 Len=0
138	4.868981794	10.200.5.2	10.200.6.112	UDP	28	53835 → 40125 Len=0
139	4.869001894	10.200.5.2	10.200.6.114	UDP	28	53835 → 40125 Len=0
140	4.869020605	10.200.5.2	10.200.6.116	UDP	28	53835 → 40125 Len=0
141	4.869034325	10.200.5.2	10.200.6.136	UDP	28	53835 → 40125 Len=0
142	4.869054766	10.200.5.2	10.200.6.139	UDP	28	53835 → 40125 Len=0
143	4.869080377	10.200.5.2	10.200.6.239	UDP	28	53835 → 40125 Len=0
144	4.869103557	10.200.5.2	10.200.6.241	UDP	28	53835 → 40125 Len=0
145	4.869115518	10.200.5.2	10.200.6.242	UDP	28	53835 → 40125 Len=0
146	4.869134478	10.200.5.2	10.200.6.250	UDP	28	53835 → 40125 Len=0
147	4.869157659	10.200.5.2	10.200.6.251	UDP	28	53835 → 40125 Len=0

Frame 130: 31... 0000 45 00 00 1c ba aa 00 00 2b 11 b4 32 0a c8 05 02 E... +...2...
Raw packet: 0010 0a c8 06 63 d2 4b 9c bd 00 08 6f e0

Masscan

On a side note, Masscan uses a similar approach to discover the available systems. However, to complete its network scan quickly, Masscan is quite aggressive in the rate at which it generates packets. The syntax is quite similar: `-p` can be followed by a port number, list, or range. Consider the following examples:

- `masscan 10.200.6.0/24 -p443`
- `masscan 10.200.6.0/24 -p80,443`
- `masscan 10.200.6.0/24 -p22-25`

Masscan is not installed on the **AttackBox**; however, it can be installed using `apt install masscan`.

Answer the questions below

Which TCP ping scan does not require a privileged account?

TCP SYN Ping

✓ Correct Answer

Which TCP ping scan requires a privileged account?

TCP ACK Ping

✓ Correct Answer

What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?

-PS23

✓ Correct Answer

